

Zeta-Equivalent Tensor Algebras of the Hénon Gassmann Twins and an Explicit Fourier Descent

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Abstract

We give an exact, bounded computation for one released Gassmann pair in the labelled action of $G = W(E_6)$, $|G| = 51840$. If H_+ and H_- are the nonconjugate subgroups of order 162 and index 320, then the three finite étale algebras

$$F_+ \otimes F_+, \quad F_+ \otimes F_-, \quad F_- \otimes F_-$$

are pairwise distinct in the Burnside ring while their rational linearizations, and hence their formal products of Dedekind zeta functions, agree. We enumerate all 36 double-coset factors, separate the 160 conjugate mixed positions into 12 factors and 8 rational field types, and give the complete 160/12/8 dictionary. A normalized Fourier decomposition of a released integral carrier has rank three and identifies an exact degree-40 mixed base through an embedded stabilizer equality. The resulting $A/B/M/F_+$ diamond has degrees 40, 80, 160, 320. Global arithmetic for all advertised fields and both retained local branches is recorded. The computation is reproducible from the released source and result inventories; it makes no claim about bad Euler factors, root numbers, automorphy, RH, or a Hilbert–Polya operator.

1 Contribution and boundary

Let $K/\mathbb{Q}$ be the released common normal closure, with $G = \text{Gal}(K/\mathbb{Q}) = W(E_6)$. The released subgroups H_+ and H_- have order 162, trivial core, equal rational permutation character, and are not conjugate. Put $F_\pm = K^{H_\pm}$ and $x = [G/H_+]$, $y = [G/H_-]$. The contribution is the exact instance

$$x^2, xy, y^2 \text{ pairwise distinct in } B(G), \quad \text{lin}(x^2) = \text{lin}(xy) = \text{lin}(y^2). \quad (1)$$

The corresponding finite étale algebras are therefore pairwise nonisomorphic despite equal rational character and zeta products. The statement is an instance-level theorem, not a new general Burnside or Fourier method; double-coset tensor factorization, Gassmann equivalence, and Artin formalism are classical (see [2, 3, 4]).

Throughout, arrays are one-based images on 27 labels, composition is left-after-right, and all arithmetic is exact. Hashes and Table-of-Marks locators are checks, never definitions. The literal scope firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.

Theorem 1.1 (Integrated C61 theorem). *For this fixed labelled $W(E_6)$ extension and released Gassmann pair, the three products x^2, xy, y^2 have the complete 36-row atlas below, are pairwise distinct in $B(G)$, and have one common rational linearization. The mixed product has 160 conjugate positions, 12 factors, and 8 rational field types. The normalized Fourier carrier has rank three, the seed-149 join is the degree-40 P_6 base, and the $A/B/M/F_+$ diamond has degrees 40, 80, 160, 320. The global table and both retained local branches are exact.*

The proof is conditional only on the released machine premises explicitly rebound in Section 8; the mathematical implications are proved in Sections 2–7. No pilot table is used as theorem authority.

2 Prior work and ownership

Gassmann's construction and Perlis's theorem supply the bridge from equal permutation characters to equal zeta functions [1, 3]. Burnside linearization and its kernel are prior infrastructure [4, 5, 6]; the double-coset tensor/compositum decomposition is also prior, with precise statements in Etienne's Lemmas 1.11–1.12 and Proposition 1.13 [7]. James's Section 3.1 collision census, especially page 22, is selection context only [2]. Kida's Proposition 4.1 and Corollary 4.2 are the closest compositum comparator, but use matched linearly disjoint extensions rather than the complete atlas inside one common $W(E_6)$ closure [8]. Accordingly, our contribution is the bounded exact conjunction in Theorem 1.1, not a general method or priority claim.

3 Tensor products from double cosets

For subgroups $H, J \leq G$, the diagonal action gives

$$(G/H) \times (G/J) \simeq \coprod_{HgJ \in H \backslash G/J} G/(H \cap gJg^{-1}). \quad (2)$$

For a representative g write $I_{ab,g} = H_a \cap gH_b g^{-1}$ and $J_{ab,g} = \langle H_a, gH_b g^{-1} \rangle$. Core-freeness and the common normal closure identify the factor and base fields as $K^{I_{ab,g}} = F_a F_b^g$ and $K^{J_{ab,g}} = F_a \cap F_b^g$.

Lemma 3.1 (Tensor and fixed-field dictionary). *For $H, J \leq G$,*

$$K^H \otimes_{\mathbb{Q}} K^J \simeq \prod_{HgJ \in H \backslash G/J} K^{H \cap gJg^{-1}}, \quad K^H K^J = K^{H \cap J}, \quad K^H \cap K^J = K^{\langle H, J \rangle}.$$

Proof. The orbit through (H, gJ) in $(G/H) \times (G/J)$ has stabilizer $H \cap gJg^{-1}$, and two such points share an orbit precisely when their representatives share a double coset. Passing from finite G -sets to finite etale algebras gives the first identity. The compositum is fixed by $H \cap J$ and the intersection by the subgroup generated by H and J , proving the remaining identities by Galois correspondence. $\square$

Lemma 3.2 (Core-free isomorphism criterion). *If $I_1, I_2 \leq G$ have trivial core, then K^{I_1} and K^{I_2} are $\mathbb{Q}$ -isomorphic if and only if I_1 and I_2 are conjugate in G .*

Proof. Trivial core makes K the normal closure of either fixed field. A $\mathbb{Q}$ -isomorphism extends to an automorphism of an algebraic closure and restricts on the common normal closure to some $g \in G$. For $\alpha \in K^{I_1}$ and $h \in I_1$, $(ghg^{-1})(g\alpha) = g(h\alpha) = g\alpha$. Thus gI_1g^{-1} fixes K^{I_2} , and equal degrees force equality. Conversely a conjugator maps one fixed field to the other. $\square$

3.1 Complete 36-row factor atlas

The following is the complete row inventory. The label Q_i is a rational isomorphism type; the listed degree is the factor degree and the multiplicity is one per double-coset row.

Table 1: All 36 tensor rows.

product	row	Q type	degree	multiplicity
T_{++}	1	Q1	320	1
T_{++}	2	Q1	320	1
T_{++}	3	Q2	960	1
T_{++}	4	Q2	960	1
T_{++}	5	Q3	1920	1

product	row	Q type	degree	multiplicity
T_{++}	6	Q4	5760	1
T_{++}	7	Q4	5760	1
T_{++}	8	Q5	8640	1
T_{++}	9	Q5	8640	1
T_{++}	10	Q6	17280	1
T_{++}	11	Q7	25920	1
T_{++}	12	Q7	25920	1
T_{+-}	1	Q8	640	1
T_{+-}	2	Q9	960	1
T_{+-}	3	Q9	960	1
T_{+-}	4	Q10	1920	1
T_{+-}	5	Q11	2880	1
T_{+-}	6	Q11	2880	1
T_{+-}	7	Q12	2880	1
T_{+-}	8	Q12	2880	1
T_{+-}	9	Q13	8640	1
T_{+-}	10	Q13	8640	1
T_{+-}	11	Q14	17280	1
T_{+-}	12	Q15	51840	1
T_{--}	1	Q16	320	1
T_{--}	2	Q16	320	1
T_{--}	3	Q17	960	1
T_{--}	4	Q17	960	1
T_{--}	5	Q3	1920	1
T_{--}	6	Q4	5760	1
T_{--}	7	Q4	5760	1
T_{--}	8	Q18	8640	1
T_{--}	9	Q18	8640	1
T_{--}	10	Q6	17280	1
T_{--}	11	Q7	25920	1
T_{--}	12	Q7	25920	1

Each product has dimension 102400. Thus the spectra are

$$T_{++} = T_{--} = [320^2, 960^2, 1920, 5760^2, 8640^2, 17280, 25920^2],$$

$$T_{+-} = [640, 960^2, 1920, 2880^4, 8640^2, 17280, 51840].$$

The complete Q atlas (order, normalizer order, automorphism order, core order) is:

type	$ Q $	$ N_G(Q) $	$ \text{Aut}_Q $	$ \text{core} $
Q1	162	324	2	1
Q2	54	1296	24	1
Q3	27	1296	48	1
Q4	9	324	36	1
Q5	6	24	4	1
Q6	3	1296	432	1
Q7	2	96	48	1
Q8	81	324	4	1
Q9	54	108	2	1
Q10	27	1296	48	1
Q11	18	36	2	1
Q12	18	108	6	1
Q13	6	36	6	1
Q14	3	216	72	1
Q15	1	51840	51840	1
Q16	162	324	2	1
Q17	54	216	4	1
Q18	6	24	4	1

3.2 The eight P types and pairwise separation

The join atlas has eight types:

type	$ P $	$ N_G(P) $	$ \text{Aut}_Q $	$ \text{core} $
P1	162	324	2	1
P2	648	1296	2	1
P3	1296	1296	1	1
P4	51840	51840	1	51840
P5	324	324	1	1
P6	1296	1296	1	1
P7	162	324	2	1
P8	648	1296	2	1

The plus-self join with digest 263f31237e6f5111f76fd3470b6936a1a314020255c22eab55cece395c2adeb5 and the transported minus-self join are conjugate, so they form one $P3$ class. The mixed Fourier join has digest 55d7f2df8abc6709489e9bf632c45d620b9b570e6a295a82ee6f941c24c2c6bc and is $P6$, not $P3$. This is the witness separating the three Burnside products; equal degree spectra alone would not suffice.

Proposition 3.3 (Burnside and zeta conclusions). *Equation (1) holds. The three finite etale algebras are pairwise nonisomorphic, while their products of Dedekind zeta functions agree.*

Proof. The mixed spectrum contains degrees 640, 2880, and 51840, absent from either self spectrum, so xy differs from x^2 and y^2 . The self spectra agree, but their two diagonal degree-320 constituents have H_+ and H_- type respectively. The released subgroups are nonconjugate and core-free; Lemma 3.2 separates these constituents, proving $x^2 \neq y^2$.

Rational linearization $\ell : B(G) \rightarrow R_{\mathbb{Q}}(G)$ is a ring homomorphism. From the released Gassmann identity $\ell(x) = \ell(y)$ we obtain $\ell(x^2) = \ell(xy) = \ell(y^2)$. In particular $x(x-y)$, $y(x-y)$, and $(x+y)(x-y)$ are explicit nonzero kernel elements. The tensor decompositions turn Burnside inequality into nonisomorphism of the finite etale algebras. Finally ζ_{KH} is the Artin L -function of $\text{Ind}_H^G \mathbf{1}$; multiplicativity over direct sums gives the three equal zeta products. This formal consequence computes no forbidden bad Euler data. $\square$

4 The mixed 160/12/8 dictionary

There are 160 conjugate positions, 12 mixed double cosets, and 8 rational field types. The complete type dictionary is shown in Table 2; “base” is the intersection degree and “factor” is the compositum degree. The degree-640 row recovers the released C60 field L over M (degrees 640

Table 2: Mixed 160/12/8 dictionary.

relative type	P	Q	raw multiplicity	base	factor
1	P5	Q8	1	160	640
2	P3	Q9	3	40	960
3	P6	Q10	3	40	1920
4	P4	Q11	9	1	2880
5	P4	Q12	9	1	2880
6	P4	Q13	27	1	8640
7	P4	Q14	27	1	17280
8	P4	Q15	81	1	51840

and 160); the degree-51840 row is the unique mixed maximum K . These uniqueness statements are restricted to the twelve mixed rows. The field dictionary follows from Galois correspondence, not from subgroup order or hash coincidence.

Proposition 4.1 (Mixed classification). *The mixed 160 positions form 12 double cosets and exactly 8 $\mathbb{Q}$ -isomorphism classes, with the multiplicities and degrees in Table 2.*

Proof. The 160 conjugates of H_- were enumerated from the released labelled arrays. Their left H_+ -orbits give 12 double cosets. Every intersection was reconstructed as a complete embedded subgroup and all eight type representatives have trivial core. Lemma 3.2 therefore converts the eight conjugacy classes into exactly eight rational field types. The raw counts satisfy $1 + 3 + 3 + 9 + 9 + 27 + 27 + 81 = 160$, and the twelve factor degrees sum to 102400. For type 1 the complete intersection and join equal the released groups J and N , so Lemma 3.1 gives L/M . Type 8 has trivial intersection and factor K . $\square$

5 Product-form resolvents and Fourier descent

The 14 source-owned carriers use a regular Lagrange basis in the 27 labelled roots. At the completely split prime $p = 692717$, every advertised orbit value is distinct both in characteristic zero and modulo p . This evaluated noncollision is what upgrades formal stabilizers to field degrees; no expanded characteristic-zero coefficient vector is claimed.

Let λ be the released primitive carrier. The exact Fourier components satisfy

$$R_0 = 0, \quad r_+ = R_+/2, \quad r_3 = R_3/4, \quad r_0 = r_+ r_3, \quad (3)$$

and the normalized reconstruction is

$$4\lambda = \text{Tr} + 2r_+ + 4r_3. \quad (4)$$

The orbit-span dimension over M is three. The identity values at p are

$$\text{Tr} = 581739, \quad r_+ = 643771, \quad r_3 = 119649, \quad r_0 = 582281 \pmod{p}.$$

The 243-term trace carrier has digest `a7398d36cea0c83ace64466a579e21666731d1e3c8e8641df4ce036c79de2bd7`. The exact square-class identity is $[\delta_0] = [\delta_+][\delta_3]$.

Set $S_+ = \text{Stab}_G(r_+)$ and $T_+ = \text{Stab}_G(\{r_+, -r_+\})$. The seed-149 mixed join equals the same embedded element set as T_+ , of order 1296; this is stronger than an abstract group isomorphism or a Table-of-Marks locator. With

$$A = \mathbb{Q}(\delta_+) = K^{T_+}, \quad B = \mathbb{Q}(r_+) = K^{S_+}, \quad M = K^N, \quad F_+ = K^{H_+},$$

the exact diamond is

$$[A : \mathbb{Q}], [B : \mathbb{Q}], [M : \mathbb{Q}], [F_+ : \mathbb{Q}] = 40, 80, 160, 320, \quad B \cap M = A, \quad BM = F_+. \quad (5)$$

Moreover $\text{Aut}_{\mathbb{Q}}(A) = 1$, $|\text{Aut}_{\mathbb{Q}}(B)| = 2$, and both routes around the diamond agree on relative discriminant norms.

Proposition 5.1 (Fourier rank and evaluated fields). *The four $V_4 = N/J$ conjugates of λ span a three-dimensional M -space, and the displayed $A/B/M/F_+$ diagram is the exact evaluated fixed-field diamond.*

Proof. For each rational character χ of V_4 , form $R_\chi = \sum_{q \in V_4} \chi(q)q(\lambda)$. Character orthogonality gives $4\lambda = \text{Tr} + R_+ + R_0 + R_3$. Exact sparse arithmetic proves $R_0 = 0$, $R_+ = 2r_+$, and $R_3 = 4r_3$, giving Equation (4). The three remaining terms lie in distinct rational eigenspaces. Their nonzero reductions at $p = 692717$ prove nonvanishing in characteristic zero, hence their independence and rank three.

If an integral formal carrier is fixed by H , it has at most $[G : H]$ characteristic-zero conjugates. When all $[G : H]$ formal orbit values have pairwise distinct reductions at a completely split prime, no two characteristic-zero values coincide. Thus the evaluated stabilizer is exactly H and the carrier generates K^H . Applying this argument to $r_+, \delta_+, r_3, \delta_3, r_0, \delta_0$ gives orbit sizes 80, 40, 320, 160, 320, 160.

The seed-149 join equals T_+ as a complete embedded element set. Hence $A = K^{T_+} = F_+ \cap F_-^{g_{149}}$. The exact subgroup identities $S_+ \cap N = H_+$ and $\langle S_+, N \rangle = T_+$ give, by Lemma 3.1, $BM = F_+$ and $B \cap M = A$. Subgroup orders give the four degrees and normalizers give the automorphism orders. In particular $\mathbb{Q}(r_+) \neq F_+$ and $\mathbb{Q}(\delta_+) \neq M$. $\square$

The product $r_0 = r_+r_3$ is distinct from the vanished direct component R_0 . Squaring yields $\delta_0 = \delta_+\delta_3$, hence $[\delta_0] = [\delta_+][\delta_3]$ in $M^\times/M^{\times 2}$.

6 Global arithmetic

All discriminants below are field discriminants obtained from permutation conductors; they are not polynomial or order discriminants. The exponent vector is ordered $(3, 5, \Pi_A, \Pi_B)$ and the signature is (r_1, r_2) .

field	degree	sign	signature	exponents
C1	160	+	(16,72)	(308,248,96,80)
C2	40	+	(8,16)	(68,62,18,20)
C3	40	-	(6,17)	(75,61,24,15)
C4	1	+	(1,0)	(0,0,0,0)
E1	640	+	(0,320)	(1264,992,384,320)
E2	960	+	(16,472)	(1944,1488,624,480)
E3	1920	+	(0,960)	(3808,2976,1152,960)
E4	2880	+	(16,1432)	(5872,4464,1872,1440)
E5	2880	+	(48,1416)	(5856,4464,1872,1440)
E6	8640	+	(48,4296)	(17640,13392,5616,4320)
E7	17280	+	(0,8640)	(35504,26784,11520,8640)
E8	51840	+	(0,25920)	(106560,80352,34560,25920)
B	80	+	(4,38)	(154,122,48,30)

The ramified support is exactly $\{3, 5\}$ together with the two certified Fourier-prime supports Π_A, Π_B . The relative norm vectors for the five diamond extensions are independently consistent along both routes.

Proposition 6.1 (Conductor calculation). *The displayed signatures and field-discriminant vectors follow from the exact permutation orbit vectors, and both diamond routes agree.*

Proof. For a degree- n action with o_∞ complex-conjugation orbits, $r_1 = 2o_\infty - n$ and $r_2 = n - o_\infty$. For the released filtration the exact exponents are

$$v_3 = (n - o_{I_3}) + \frac{1}{2}(n - o_{P_3}) + (n - o_{Q_3}), \quad v_5 = (n - o_{I_5}) + \frac{3}{4}(n - o_{P_5}),$$

$$v_{\Pi_A} = n - o_{C_3}, \quad v_{\Pi_B} = n - o_{C_2}.$$

Substitution of the independently reconstructed orbit vectors gives every row above. For E/C , the tower formula

$$\text{Disc}(E/\mathbb{Q}) = \text{Disc}(C/\mathbb{Q})^{[E:C]} N_{C/\mathbb{Q}}(\mathfrak{d}_{E/C})$$

gives

$$N(\mathfrak{d}_{B/A}) = (4, 0, 0, 0), \quad N(\mathfrak{d}_{M/A}) = (8, 4, 0, 20),$$

$$N(\mathfrak{d}_{F_+/B}) = (8, 8, 0, 40), \quad N(\mathfrak{d}_{F_+/A}) = (24, 8, 0, 40), \quad N(\mathfrak{d}_{F_+/M}) = (8, 0, 0, 0).$$

Both $2(8, 4, 0, 20) + (8, 0, 0, 0)$ and $4(4, 0, 0, 0) + (8, 8, 0, 40)$ equal $(24, 8, 0, 40)$, proving route consistency. $\square$

7 Both local branches and complementarity

We retain, without selecting, the two released Table-of-Marks branches $D_3 = \text{ToM } 140$ and $D_3 = \text{ToM } 206$. The table gives factor counts, degree totals, and different totals for the bases B, C_1, C_2, C_3, C_4 and the eight mixed factors E_i .

field	ToM 140			ToM 206		
	factors	degree	different	factors	degree	different
B	10	80	154	8	80	154
C1	22	160	308	11	160	308
C2	10	40	68	5	40	68
C3	7	40	75	5	40	75
C4	1	1	0	1	1	0
E1	56	640	1264	28	640	1264
E2	72	960	1944	36	960	1944
E3	160	1920	3808	80	1920	3808
E4	188	2880	5872	94	2880	5872
E5	204	2880	5856	102	2880	5856
E6	552	8640	17640	276	8640	17640
E7	968	17280	35504	484	17280	35504
E8	2880	51840	106560	1440	51840	106560

Every uncollected row was checked, every ramified relative row is tame with $(e, d) = (2, 1)$, and the primewise laws are

$$\gcd(\mathfrak{d}_{F_+/M}, \mathfrak{d}_{F_3/M}) = 1, \quad \mathfrak{d}_{F_0/M} = \mathfrak{d}_{F_+/M} \mathfrak{d}_{F_3/M}, \quad \mathfrak{d}_{L/M} = \mathfrak{d}_{F_0/M}^2.$$

The archimedean calculation gives complementary real splitting and $\mathbb{C} \times \mathbb{C}$ over every real place of M . The tuples (n, e, f, d) do not classify local fields.

Proof of the local assertion in Theorem 1.1. For each base prime and relative row (g, e, f, d) , set $a = gfd$. Every uncollected row in both branches verifies

$$\min(a_+, a_3) = 0, \quad a_0 = a_+ + a_3, \quad qquad a_L = 2a_0.$$

The valuation identities imply the three displayed ideal equalities. The ToM 140 and 206 branch populations are respectively $(8, 8, 6, 0)$ and $(4, 4, 3, 0)$ for $(H_+, H_3, 1, H_0)$, while both have residue-degree masses $(8, 8, 6, 0)$; no branch is selected. Every ramified relative row has $e = 2, d = 1$, hence is tame. Independently, the infinite types $H_+ : 8, H_3 : 8, H_0 : 0$ prove the archimedean complementarity. $\square$

8 Reproducibility and evidence

The official clean refresh and mandatory replay each passed 57/57 using GAP 4.11.1 and TomLib 1.2.9 for the independent group lane [9, 10]. The authoritative payload, certificate, report, schema, and scoped manifest have digests

payload	b7fb70451433fd4c93fd9d60a338426362f42c4594ff4de5a35e25f49819ab1a
certificate	c6d64787e91c78e7e7f74b88720ebf169c13d8200f78c585abe63d4e1b9b2dec
report	4b8b3bd21209cc9346ac7e38fbb9771d0b8b33de0cd28ccbb117beaa95e8c161
schema (pretty)	a446d693742a327dd0b1edd3ad5da97a50dc76cc0f1f7e171096df341ad10e76
manifest	eb48ffb2ec728ca9d1966cfd22b6a9f2eba447bcf872d2068f4604112d19a6d1
producer	dadf8899f2fe82b65131a43ffbe438602db79a12654489b34d35ae8a6ee83d99
checker	571de05ce06cf98c1acb6809800cf5f212755ce91c5d2f8eb5733eb1aa708887

The payload has 19078 scalar leaves and the schema 29; the checker rejects 19109 value, 19109 type, and 14 structural certificate mutations, plus 6 group, 4 resolver, and 2 artifact contract rebounds. All 30 scope leaves are false. The paper uses no external pilot as theorem authority and no generated certificate as a definition.

9 Limitations and conclusion

Equation (1) is the exact bounded conclusion. We do not claim bad Euler data, decomposition Frobenius, epsilon or root numbers, Artin holomorphy, automorphy, analytic continuation, RH, a Hilbert–Polya operator, maximal orders, integral bases, class numbers, regulators, local-field classification, rational points, or branch selection. Both local branches remain part of the theorem data. Release and promotion are workflow states distinct from the mathematical certificate; this manuscript reports the machine-complete evidence while retaining that separation.

A Claim-to-evidence matrix

claim	evidence
36 tensor rows and characters	G1 payload and independent GAP projection
160/12/8 dictionary	G2 mixed resolver rows, core-free extension checks
field degrees	G3 product-form carriers and complete split-prime noncollision
Fourier bridge and diamond	G4 exact sparse identities, stabilizers, embedded seed-149 equality
global arithmetic	G5 conductor and relative-norm vectors
both local branches	G6 complete uncollected rows and primewise ideal laws
independence and firewall	G7 source separation, mutation suites, 30 false scope leaves

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